Invention Disclosure Form (IDF) – Part B: Technical Disclosure (Draft)

NeuroSonic project · Allwin Raj · Vellore Institute of Technology · 3 November 2026

> **Note on format.** This draft follows the usual structure of an Invention Disclosure Form (Part B: technical disclosure). If the institute's IPR cell provides its own IDF-B template, the text below can be transferred section by section. Highlighted fields must be completed by the inventor(s). **Do not publish the paper drafts (DA1–DA3, Lab 10) or the demonstration video publicly before the IPR cell has advised on filing**; public disclosure before filing can destroy novelty in many jurisdictions.

1. Title of the Invention

System and method for camera-motion-compensated spoken walking-path event alerts, using an unmodified smartphone browser as a sensing thin client and a nearby offline inference host, for pedestrians with visual impairment.

2. Inventor(s) and Applicant

Name	Affiliation	Contact	Contribution
Allwin Raj	[School / Department, Register no.], VIT Vellore	theallwinraj@gmail.com	[%]
[Guide / co-inventor, if any]	[ ]	[ ]	[%]

Applicant: [Vellore Institute of Technology, per institute IPR policy]. Funding: [none / scheme].

3. Field of the Invention

Assistive technology for persons with visual impairment; real-time computer vision (object detection, multi-object tracking, ego-motion estimation); human–computer interaction by speech; pedestrian navigation. Suggested classification areas for the search (to be confirmed by the IPR cell): A61H 3/06 (walking aids for blind persons), G06T 7/246 (motion analysis using feature-based methods), G01C 21/36 (route guidance input/output), G10L 13/00 (speech synthesis), G08B 21/02 (alarms for ensuring the safety of persons).

4. Background and Problem

At least 2.2 billion people have a vision impairment and 43.3 million were blind in 2020 [1], [2]. Electronic travel aids either measure range (ultrasonic canes [3], [4]), recognise objects on demand (phone apps [5]), or use dedicated depth wearables, robots or garments [6], [7], [8]. A camera carried by a walking person moves continuously; in the image, every static object therefore appears to move. Systems that detect objects frame by frame cannot tell a person stepping into the user's way from a parked car that merely appears to move because the user turns. Warning about all apparent motion produces false alarms; warning about none misses the most important events. Dedicated hardware is expensive, and cloud services need connectivity and send images off the device.

5. Prior Art Known to the Inventor

Document	Disclosure	Difference from the present invention
US 4,280,204 [3]	ultrasonic cane, two range modes	no recognition, no motion semantics
US 2005/0208457 A1 [9]	head-mounted camera, object classification on request, speech	single images on request; no tracking, no ego-motion compensation, no path events
US 9,429,446 B1 [10]	hand-held device; camera/sonar; path planning; spoken directions	no background-relative motion; no suppression during camera motion
US 9,846,231 B2 [11]	wearable ultrasound gated by gait phase (accelerometer)	different sensor; gating by foot contact, not by image ego-motion
US 10,299,982 B2 [12]	phone camera builds a 3-D room model, touch exploration	static model; not live walking alerts
US 10,809,079 B2 [8]	vest with GPS, cameras, ultrasonic; static/dynamic obstacles for routing	dedicated garment; no disclosure (in the reviewed text) of image ego-motion-compensated entering/crossing events from a phone browser client
US 11,318,050 B2 [13]	phone location sensors; direction-filtered narration	no camera perception
Gao et al., WOAD [6]	NIR + ToF glasses, FPGA, phone detector, audio/haptic alerts	custom sensor and glasses; frame-wise detection in the released code
ByteTrack [14]; LK/RANSAC [15], [16]	general tracking and motion estimation methods	general methods; not their application as claimed here

The search was limited to US documents on Google Patents and to the publications reviewed in DA1; it is not a professional search. Indian (InPASS) and WIPO (PATENTSCOPE) searches are

still required.

6. Summary of the Invention

A smartphone runs only a web page. It captures camera frames at a rate set by the host (for example every 200 ms when a GPU detector is available, 400 ms otherwise), keeps at most one frame in flight, and plays the speech the host returns. A nearby host on the same local network, without internet, (i) detects obstacle-class objects; (ii) estimates the camera's own motion from background feature points outside the detected objects; (iii) predicts each tracked object's position under that camera motion and associates detections in two confidence stages; (iv) stores each object's displacement *relative to the background*; (v) maintains a confirmed walking-path state for a lower-central image region; and (vi) emits a spoken "entering" or "crossing" event only when a confirmed track of a person, vehicle or animal moves into or across the walking-path region faster than a threshold relative to the background, while suppressing all events during strong or unknown camera motion and in uncertain or blocked views. All messages come from a fixed vocabulary that never states distance or safety.

7. Detailed Description

7.1 Architecture

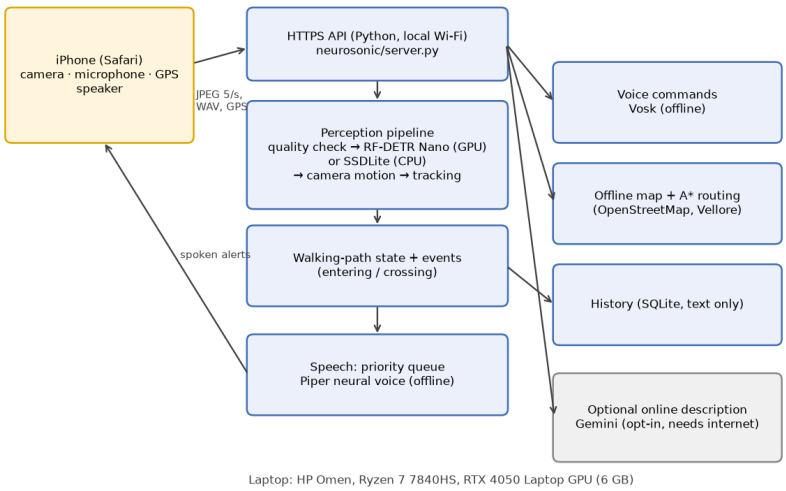


Fig. 1. System architecture (phone thin client and local host).

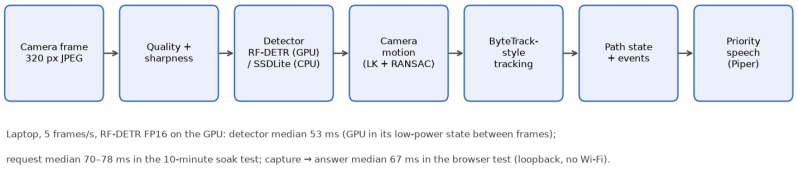


Fig. 2. Per-frame processing on the host.

7.2 Ego-motion estimation

Up to 150 corner points are selected on a reduced grey image outside the detected boxes, tracked into the next frame by pyramidal Lucas–Kanade flow, and a similarity transform M (translation, scale, rotation) is fitted by RANSAC; if fewer than 12 points or inliers remain, the motion is marked unknown.

7.3 Motion-compensated association and background-relative motion

Each track's previous box is transformed by M to predict its current position. Detections of confidence ≥ 0.55 are matched first by IoU within each class; detections of confidence 0.40–0.55 may only extend existing tracks; tracks end after three missed frames. For each update, the background-relative displacement $\Delta_{rel} = c_t - M(c_{t-1})$ of the box centre c is stored.

7.4 Event rules and suppression

Entering: a confirmed track (≥ 3 updates over ≥ 0.4 s) that was outside the walking-path region at the start of a 1.5 s window, overlaps it now and moves towards it at ≥ 0.12 image widths per second relative to the background. *Crossing:* the centre passes from below 35 % to above 65 % of the width (or back) within 3 s at ≥ 0.15 widths/s with the box bottom in the lower half. *Suppression:* no event while the camera shifts more than 6 % of the width per frame, zooms more than 4 %, or its motion is unknown; none in blocked, uncertain or unavailable views; one event per track; events at least 3 s apart.

7.5 Thin client and offline host

The phone uses only a standard browser over HTTPS with a locally generated certificate authority (required by the browser for camera and microphone access). The host also performs offline speech synthesis, offline command recognition and offline route planning; images are processed in memory and never stored.

8. Features Believed to Be New (for the IPR Cell's Assessment)

1. Announcing an object as *entering* or *crossing* a walking-path image region on the basis of its *background-relative* velocity, computed from ego-motion estimated on feature points excluded from detected objects.
2. Combining that rule with suppression of announcements when the estimated ego-motion exceeds translation or scale limits or is unknown, and with confirmed walking-path states.
3. Two-stage confidence association in which the prediction step is the estimated camera transform rather than a motion model of the object.
4. A thin-client arrangement in which an unmodified smartphone browser supplies frames at a host-selected interval with a single frame in flight, and the host returns speech from a fixed, distance-free vocabulary, all without internet.

Whether these features, alone or in combination, are novel and inventive can only be decided after a professional search. General methods used (Lucas–Kanade flow, RANSAC, IoU association, transformer detectors) are prior art in themselves and are not claimed as such.

9. Draft Claims (for discussion only)

1. A method for alerting a pedestrian, comprising: receiving successive images from a camera carried by the pedestrian; detecting objects of predetermined classes in each image; estimating a geometric transform of the camera between successive images from feature points located outside the detected objects; predicting the position of each previously tracked object by applying the transform; associating detections with predicted tracks; computing for each track a displacement relative to the background as the difference between the observed position and the position predicted by the transform; and generating a spoken alert that the object is entering or crossing a predefined walking-path region of the image when the background-relative displacement rate towards or across the region exceeds a threshold.
2. The method of claim 1, wherein no alert is generated while the estimated transform exceeds a translation or scale limit, or cannot be estimated.
3. The method of claim 1, wherein association is performed first for detections above a first confidence and then, for remaining tracks only, for detections between a second, lower confidence and the first.
4. The method of claim 1, wherein an alert is generated at most once per track and at least a minimum interval after the previous alert.
5. The method of claim 1, wherein the images are captured by a web browser of a smartphone and sent over a local network to a host that performs the detecting, estimating, associating and generating without access to the internet, and returns synthesised speech to the browser.
6. The method of claim 1, wherein the spoken alerts are selected from a vocabulary that contains no distance values and no statement that the path is safe.
7. A system comprising a smartphone running a web client and a host computer configured to perform the method of any of claims 1 to 6.

10. Advantages

- Uses hardware many users already have (a phone and a laptop) instead of custom wearables costing hundreds to thousands of US dollars [17].
- Reduces false alarms from the user's own movement while still announcing objects that move into the way.
- Works offline; images never leave the local network unless the user opts in to cloud description.
- Honest wording reduces the risk of over-reliance.

11. Reduction to Practice and Evidence

Implemented and tested on a laptop (RTX 4050 Laptop GPU): RF-DETR Nano detector (person AP@0.5 0.660, recall 0.729 on 365 near-IR images); about 5 frames/s with a median capture-to-answer time of 67 ms in browser tests; ten-minute soak test with 2980 frames and 0 errors; 22 perception tests including no event for a static car during a synthetic pan and correct entering and crossing events. Physical tests with an iPhone and with users are **pending**. Technology readiness: laboratory validation (approximately TRL 4).

12. Applications and Commercial Potential

Supplementary mobility aid for people with low vision or blindness; training aid for orientation and mobility instructors; awareness module for low-cost robots or carts. A future native phone app with on-device inference would remove the laptop. Potential partners: NGOs and rehabilitation centres for the visually impaired, assistive-technology manufacturers.

13. Disclosure Status and Third-Party Rights

Question	Answer
Published or presented publicly?	[Confirm: course reviews only / any public posting, GitHub, video, conference?]
Planned publication date	[ ] (file first)
Third-party code and models	RF-DETR (Apache-2.0), OpenCV (Apache-2.0), PyTorch (BSD), Piper (GPL-3.0+; voice AGPL-3.0), Vosk (Apache-2.0), OpenStreetMap (ODbL); base-paper code used only as a research baseline
Agreements / funding obligations	[ ]

14. Declaration

I declare that the information given above is true to the best of my knowledge.

Signature of inventor(s) Date: [ ]
Allwin Raj

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